Analysis & Study of Complex Network Topologies

Dionysios Chontzakis

This report summarizes the study of five canonical synthetic network families — regular, random, random geometric, scale free, and small world — and the comparison of them with a real network from *Game of Thrones* Season 1.

This work combines visual inspection, centrality distributions, graph-level metrics, a degree-rank diagnostic, and community detection to position the real graph relative to idealized baselines.

Contents

1. Synthetic baselines used in the notebook

The visualization of the five benchmark familiew provide the vocabulary for the real-network classification: regular graphs emphasize uniformity, random graphs short paths, geometric graphs spatial locality, scale free graphs hubs, and small-world graphs clustering plus short routes.

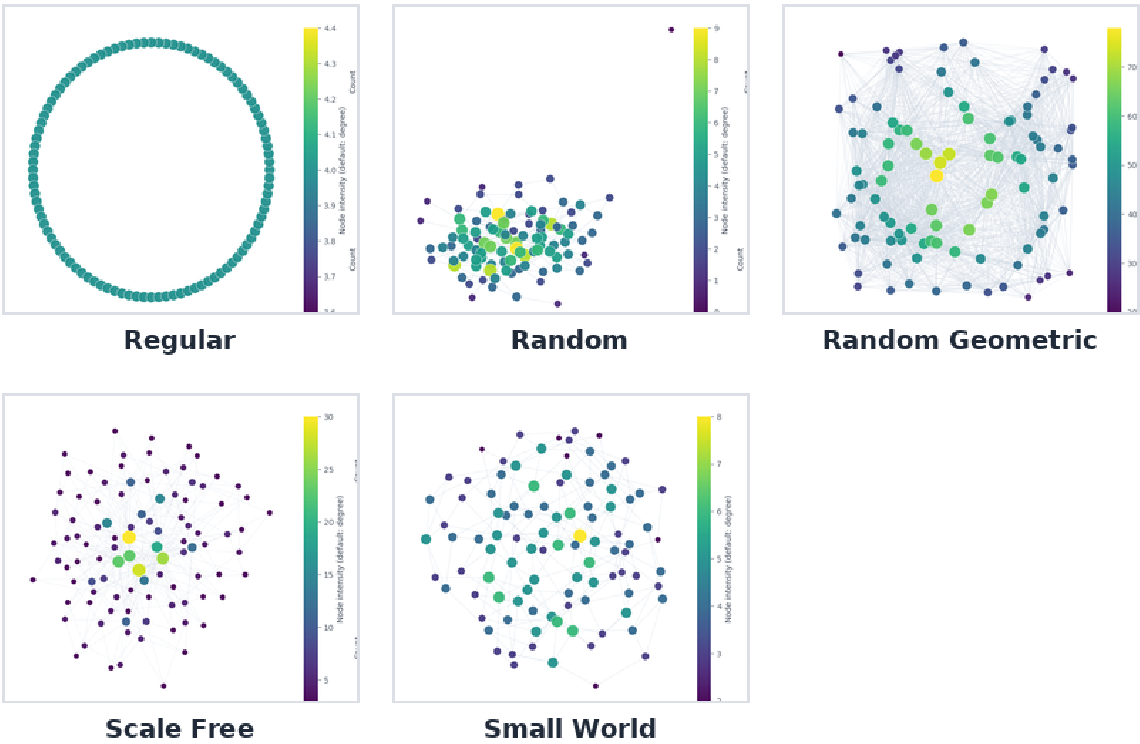


Figure 1: From left to right and top to bottom: Regular, Random, Random Geometric, Scale Free, and Small World.

Model signatures. Regular graphs have no degree variability and long routes; random graphs lose clustering but keep paths short; geometric graphs create dense local neighborhoods; scale-free graphs develop heavy right tails because hubs attract edges; and small-world graphs retain clustering while still allowing fast global communication.

2. Real network overview: *Game of Thrones* Season 1

The real-data section loads a weighted character-interaction network with 126 nodes and 549 edges. The dashboard below combines a weighted network view with distributions of degree distribution, clustering coefficient, betweenness centrality, and closeness centrality, plus highlighted subgraphs for the most central characters.

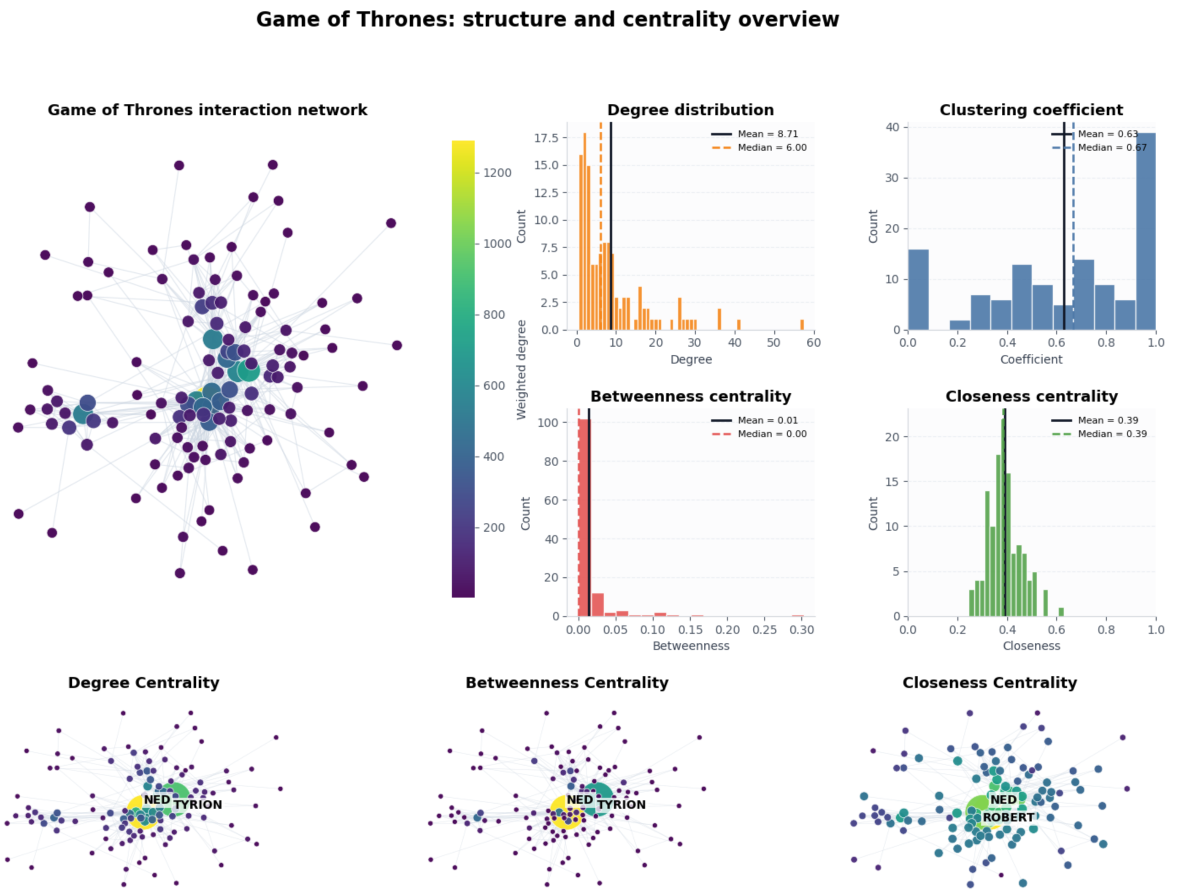


Figure 2: Dashboard for the *Game of Thrones* network. Larger and brighter nodes correspond to characters with stronger weighted interaction totals; the side histograms summarize the main centrality and clustering distributions; the botttom graphs highlight the most central characters based on that metric.

Reading the dashboard. The graph is visibly heterogeneous: a compact narrative core is surrounded by less-connected peripheral characters. The degree mean is 8.71, the clustering coefficient mean is 0.63, the betweenness cetrality mean is 0.01, the closeness cetrality mean is 0.39 and the average shortest path is only 2.64.

Centrality Table

node	degree_centrality	betweenness	closeness	eigenvector
NED	0.4560	0.3033	0.6281	0.3151
TYRION	0.3280	0.1630	0.5435	0.2298
CATELYN	0.2880	0.1183	0.5507	0.2369
ROBERT	0.2880	0.1104	0.5531	0.2482
ROBB	0.2400	0.0800	0.5123	0.2071
CERSEI	0.2320	0.0177	0.5081	0.2393
ARYA	0.2240	0.0660	0.5020	0.2072
JOFFREY	0.2160	0.0286	0.5102	0.2195
JON	0.2080	0.0853	0.5187	0.1683
SANSA	0.2080	0.0201	0.4921	0.2026

Figure 3: Centrality table for the *Game of Thrones* network. Here, the centrality metrics present the most central charecters.

Reading the table. NED dominates the centrality rankings, while TYRION and ROBERT also recur.

3. Distributional comparison against synthetic models

This work directly overlays the *Game of Thrones* distributions against one representative instance from each synthetic family. This is useful because it tests whether the real graph behaves like each model across degree, clustering, betweenness, and closeness at the same time.

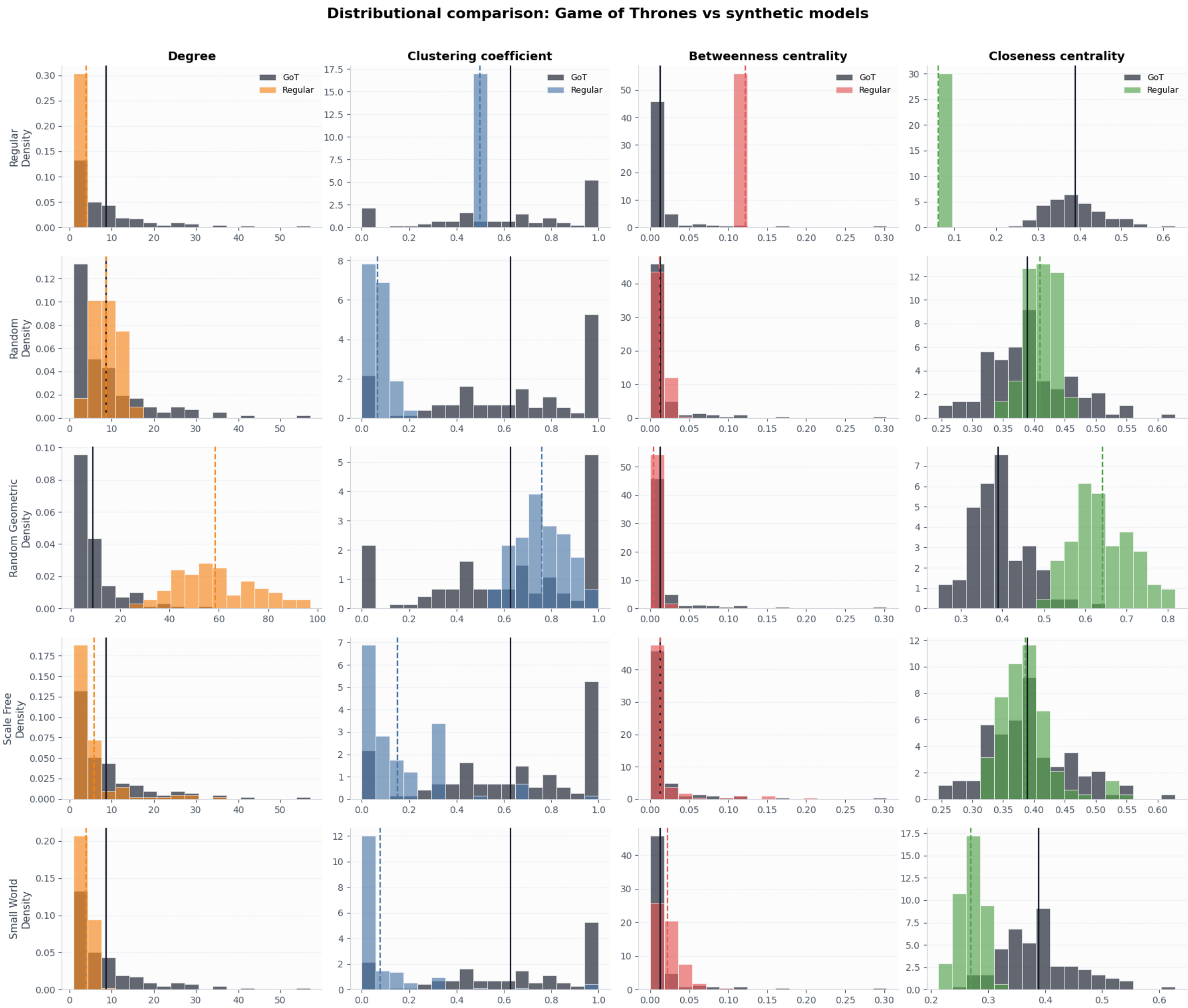
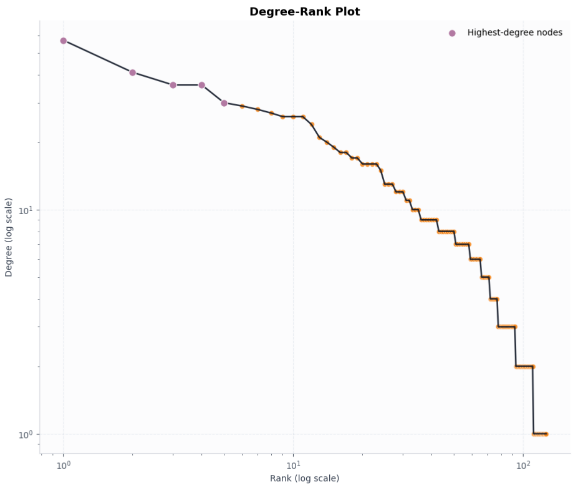


Figure 4: Distributional comparison of *Game of Thrones* versus synthetic baselines.

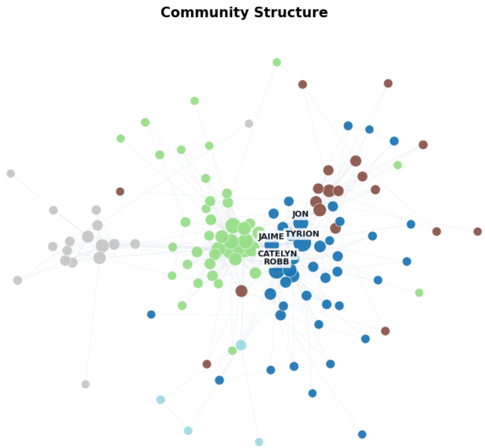
Main interpretation. The real network is not regular because its degree spread is far too heterogeneous. It is also not purely random because its clustering is much higher than the random baseline. The geometric model reproduces strong clustering, but it implies a much more spatial and assortative structure than the real data. From the last 2 synthetic networks, the scale free is overall the best description. That conclusion comes both from the node degree distribution as well as from the centrality metrics.

4. Hub evidence and community structure

Two additional figures sharpen the classification. The degree-rank plot checks for a hub-dominated tail, while the community plot shows whether the network contains mesoscopic groups rather than being homogeneous throughout.



(a) Degree-rank plot of the real network.



(b) Greedy modularity communities detected on the giant component.

Why these figures matter. The degree-rank curve decays gradually rather than collapsing immediately, which is consistent with hub-oriented, heavy-tailed structure. The community analysis detects 5 communities with modularity 0.4685, confirming that the network is socially organized into internally denser groups. In substantive terms, the network is both centralized and clustered: a few pivotal characters bridge the story world, but there are still recognizable narrative communities. That combination is typical of real interaction networks.